

Baseline VAR – Canada GDP, Housing & Inflation

0. Packages

```
R_packages <- c("readr", "dplyr", "tseries", "vars", "ggplot2", "lubridate", "scales", "tsDyn", "xts", "

options(repos = c(CRAN="http://cran.rstudio.com"))
if (!requireNamespace("librarian", quietly = TRUE)) install.packages("librarian")
librarian::shelf(R_packages)

if (!requireNamespace("tinytex", quietly = TRUE)) install.packages("tinytex")
if (Sys.which("pdflatex")=="") {library(tinytex); tinytex::install_tinytex()}
```

1. Data

```
df <- read_csv("balanced_can_md.csv", show_col_types = FALSE)

df_subset <- df %>%
  dplyr::select(
    Date,
    GDP_new,
    CPI_ALL_CAN,
    NHOUSE_P_CAN,
    EMP_CAN
  )

head(df_subset)
```

```
## # A tibble: 6 x 5
##   Date      GDP_new CPI_ALL_CAN NHOUSE_P_CAN EMP_CAN
##   <date>      <dbl>      <dbl>      <dbl>      <dbl>
## 1 1981-01-01  0.00127      0.0150      0.00168  0.00595
## 2 1981-02-01  0.00304      0.0106      0.0130   0.00683
## 3 1981-03-01 -0.000820     0.0126      0.0154 -0.000408
## 4 1981-04-01  0.0118      0.00623     0.0152   0.00328
## 5 1981-05-01 -0.00836      0.00825     0.0174   0.00265
## 6 1981-06-01 -0.00148      0.0163      0.00491  0.00262
```

1.1 Variable: National-level variables from Fortin

```
## initial screening out provincial variables and some duplicate one.
```

```
vars_all_blocks <- c(
  "GDP_new", "IP_new", "NDM_new", "DM_new", "CON_new", "RT_new", "WT_new",
  "PA_new", "FIN_new", "BSI_new", "GPI_new", "SPI_new", "TOT_HRS_CAN",
  "CPI_ALL_CAN", "IPPI_MACH_CAN", "IPPI_METAL_CAN",
  "OILP_new", "OIL_CAN_new",
  "BANK_RATE_L", "TBILL_3M", "TBILL_6M",
  "G_AVG_5.10.Bank_rate", "G_AVG_10p.TBILL_3M",
  "USDCAD_new", "GBPCAD_new", "OILP_new", "OIL_CAN_new", "WTISPLC",
  "Exp_BP_new", "Imp_BP_new", "FOR_SEC_NETFLOW",
  "hstart_CAN_new", "NHOUSE_P_CAN", "CRED_HOUS_MORT", "CRED_HOUS",
  "FIN_new", "EMP_CONS_CAN", "EMP_FOR_OIL_CAN", "TSX_CLO", "TSX_HI",
  "EMP_CAN", "EMP_SERV_CAN", "EMP_FIN_CAN", "EMP_MANU_CAN",
  "EMP_CONS_CAN", "EMP_PART_CAN", "EMP_FOR_OIL_CAN", "EMP_SALES_CAN",
  "UNEMP_CAN", "UNEMP_DURA_1.4_CAN", "UNEMP_DURA_5.13_CAN",
  "UNEMP_DURAvg_CAN_new", "CLAIMS_CAN"
)
```

```
df_large <- df %>% dplyr::select(Date, all_of(vars_all_blocks))
head(df_large)
```

```
## # A tibble: 6 x 49
##   Date      GDP_new  IP_new  NDM_new  DM_new  CON_new  RT_new  WT_new
##   <date>      <dbl>  <dbl>  <dbl>   <dbl>  <dbl>   <dbl>  <dbl>
## 1 1981-01-01  0.00127 -0.00223 -0.00355 -0.000392 0.00217 0.00151 -0.000671
## 2 1981-02-01  0.00304  0.00422  0.00172  0.0129  -0.00147 0.000579 -0.00108
## 3 1981-03-01 -0.000820  0        0.0112  -0.00277 -0.00744 0.0114  -0.00796
## 4 1981-04-01  0.0118  0.0189  0.0156  0.0135  0.0169  0.0259  0.000353
## 5 1981-05-01 -0.00836 -0.0176 -0.0173 -0.00398 0.00903 -0.0475 -0.0124
## 6 1981-06-01 -0.00148  0.00472 -0.00412  0.0111  0.00146 -0.00309  0.00184
## # i 41 more variables: PA_new <dbl>, FIN_new <dbl>, BSI_new <dbl>,
## #   GPI_new <dbl>, SPI_new <dbl>, TOT_HRS_CAN <dbl>, CPI_ALL_CAN <dbl>,
## #   IPPI_MACH_CAN <dbl>, IPPI_METAL_CAN <dbl>, OILP_new <dbl>,
## #   OIL_CAN_new <dbl>, BANK_RATE_L <dbl>, TBILL_3M <dbl>, TBILL_6M <dbl>,
## #   G_AVG_5.10.Bank_rate <dbl>, G_AVG_10p.TBILL_3M <dbl>, USDCAD_new <dbl>,
## #   GBPCAD_new <dbl>, WTISPLC <dbl>, Exp_BP_new <dbl>, Imp_BP_new <dbl>,
## #   FOR_SEC_NETFLOW <dbl>, hstart_CAN_new <dbl>, NHOUSE_P_CAN <dbl>, ...
```

1.2 STATIONARITY TEST

```
vars <- names(df_large)
results <- data.frame(
  variable = character(),
  statistic = numeric(),
  p_value = numeric(),
  stringsAsFactors = FALSE
)
```

```
for (v in vars) {
  x <- df_large[[v]]
```

```

if (is.numeric(x)) {

  test <- withCallingHandlers(
    adf.test(na.omit(x)),
    warning = function(w) {
      if (grepl("p-value smaller than printed p-value", w$message)) {
        invokeRestart("muffleWarning")
      }
    }
  )

  results <- rbind(
    results,
    data.frame(
      variable = v,
      statistic = as.numeric(test$statistic),
      p_value = as.numeric(test$p.value)
    )
  )
}
}

print(results)

```

```

##           variable statistic  p_value
## 1      GDP_new -7.921044 0.01000000
## 2      IP_new -7.148473 0.01000000
## 3     NDM_new -7.520620 0.01000000
## 4      DM_new -7.377472 0.01000000
## 5     CON_new -7.884777 0.01000000
## 6      RT_new -7.549817 0.01000000
## 7      WT_new -8.528149 0.01000000
## 8      PA_new -6.727507 0.01000000
## 9     FIN_new -6.987417 0.01000000
## 10     BSI_new -7.558550 0.01000000
## 11     GPI_new -7.428840 0.01000000
## 12     SPI_new -8.647956 0.01000000
## 13    TOT_HRS_CAN -9.373414 0.01000000
## 14    CPI_ALL_CAN -6.956187 0.01000000
## 15    IPPI_MACH_CAN -7.127934 0.01000000
## 16    IPPI_METAL_CAN -5.899859 0.01000000
## 17     OILP_new -8.418132 0.01000000
## 18    OIL_CAN_new -12.397439 0.01000000
## 19    BANK_RATE_L -6.682807 0.01000000
## 20     TBILL_3M -6.473829 0.01000000
## 21     TBILL_6M -6.928916 0.01000000
## 22 G_AVG_5.10.Bank_rate -4.068091 0.01000000
## 23  G_AVG_10p.TBILL_3M -3.862670 0.01586616
## 24     USDCAD_new -7.818040 0.01000000
## 25     GBPCAD_new -9.077306 0.01000000
## 26     WTISPLC -8.469651 0.01000000
## 27     Exp_BP_new -8.520924 0.01000000
## 28     Imp_BP_new -7.745129 0.01000000

```

```
## 29      FOR_SEC_NETFLOW -5.428624 0.01000000
## 30      hstart_CAN_new -9.650499 0.01000000
## 31      NHOUSE_P_CAN -4.122804 0.01000000
## 32      CRED_HOUS_MORT -3.509770 0.04136784
## 33      CRED_HOUS -3.221418 0.08423760
## 34      EMP_CONS_CAN -8.094010 0.01000000
## 35      EMP_FOR_OIL_CAN -7.127229 0.01000000
## 36      TSX_CLO -7.944596 0.01000000
## 37      TSX_HI -7.484490 0.01000000
## 38      EMP_CAN -8.090634 0.01000000
## 39      EMP_SERV_CAN -8.703637 0.01000000
## 40      EMP_FIN_CAN -8.616629 0.01000000
## 41      EMP_MANU_CAN -6.491513 0.01000000
## 42      EMP_PART_CAN -10.550797 0.01000000
## 43      EMP_SALES_CAN -8.313382 0.01000000
## 44      UNEMP_CAN -7.101288 0.01000000
## 45      UNEMP_DURA_1.4_CAN -9.425912 0.01000000
## 46      UNEMP_DURA_5.13_CAN -8.555865 0.01000000
## 47      UNEMP_DURAvg_CAN_new -7.496587 0.01000000
## 48      CLAIMS_CAN -5.772481 0.01000000
```

1.3 Construct GDP per worker growth rate estimate

```
df_subset$GDP_per_w = df_subset$GDP_new - df_subset$EMP_CAN
adf.test(df_subset$GDP_per_w)
```

```
## Warning in adf.test(df_subset$GDP_per_w): p-value smaller than printed p-value
```

```
##
## Augmented Dickey-Fuller Test
##
## data: df_subset$GDP_per_w
## Dickey-Fuller = -7.2353, Lag order = 8, p-value = 0.01
## alternative hypothesis: stationary
```

```
df_large$GDP_per_w = df_large$GDP_new - df_large$EMP_CAN
```

2. Three variable baseline

```
df_subset$Date <- as.Date(df_subset$Date)

start_year <- as.numeric(format(min(df_subset$Date), "%Y"))
start_month <- as.numeric(format(min(df_subset$Date), "%m"))
vars_3 <- c("GDP_per_w", "NHOUSE_P_CAN", "CPI_ALL_CAN")
```

2.1 K-Fold Expanding Window Cross Validation

```
X <- df_subset %>%
  dplyr::select(all_of(vars_3)) %>%
  as.matrix() %>%
  stats::na.omit()

n <- nrow(X)
## define a CV function for our purpose: optimize forecast for h= 1-3 and for h= 1-12, respectively
ts_cv_rmse_for_p <- function(X, p, K, max_h) {
  n <- nrow(X)
  if (n <= p + K + 10) return(NA_real_)

  cutpoints <- floor(seq(from = p + 20, to = n - max_h, length.out = K))

  err_list <- vector("list", max_h)

  for (t0 in cutpoints) {
    train <- X[1:t0, , drop = FALSE]

    model <- try(VAR(train, p = p, type = "const"), silent = TRUE)
    if (inherits(model, "try-error")) next

    # 1 to max_h-step-ahead forecasts from t0
    fc_all <- predict(model, n.ahead = max_h)$fcst

    for (h in 1:max_h) {
      # forecast at horizon h for all variables
      fc_h <- sapply(fc_all, function(m) m[h, "fcst"])

      # actual at t0 + h
      actual_h <- X[t0 + h, ]

      err_list[[h]] <- rbind(err_list[[h]], actual_h - fc_h)
    }
  }

  # if no errors were collected, return NA
  if (all(vapply(err_list, is.null, logical(1L)))) return(NA_real_)

  # RMSE per horizon (averaged across variables),
  rmse_h <- numeric(max_h)
  for (h in 1:max_h) {
    if (!is.null(err_list[[h]]) && nrow(err_list[[h]]) > 0L) {
      rmse_vec <- sqrt(colMeans(err_list[[h]]^2, na.rm = TRUE))
      rmse_h[h] <- mean(rmse_vec)
    } else {
      rmse_h[h] <- NA_real_
    }
  }

  mean(rmse_h[is.finite(rmse_h)])
}
```

2.2 Small VAR: Short Term Model (h=1-3)

```
min_p <- 1
max_p <- 12
step_p <- 1

p_grid <- seq(min_p, max_p, by = step_p)

results_p <- data.frame(p = integer(), avg_RMSE = numeric())

for (p in p_grid) {
  avg_rmse <- ts_cv_rmse_for_p(X, p, K = 10, max_h = 3) # CV over horizons 1-3
  results_p <- rbind(results_p, data.frame(p = p, avg_RMSE = avg_rmse))
  cat("Done CV: p =", p, "avg_RMSE =", round(avg_rmse, 6), "\n")
}
```

```
## Done CV: p = 1 avg_RMSE = 0.004199
## Done CV: p = 2 avg_RMSE = 0.004237
## Done CV: p = 3 avg_RMSE = 0.004218
## Done CV: p = 4 avg_RMSE = 0.004154
## Done CV: p = 5 avg_RMSE = 0.004339
## Done CV: p = 6 avg_RMSE = 0.004244
## Done CV: p = 7 avg_RMSE = 0.003892
## Done CV: p = 8 avg_RMSE = 0.004659
## Done CV: p = 9 avg_RMSE = 0.004638
## Done CV: p = 10 avg_RMSE = 0.004816
## Done CV: p = 11 avg_RMSE = 0.004833
## Done CV: p = 12 avg_RMSE = 0.00484
```

```
print(results_p)
```

```
##      p      avg_RMSE
## 1     1 0.004198882
## 2     2 0.004237458
## 3     3 0.004218381
## 4     4 0.004153963
## 5     5 0.004339274
## 6     6 0.004244023
## 7     7 0.003891943
## 8     8 0.004659286
## 9     9 0.004638109
## 10    10 0.004816181
## 11    11 0.004833002
## 12    12 0.004840447
```

```
valid      <- which(is.finite(results_p$avg_RMSE))
best_id    <- valid[ which.min(results_p$avg_RMSE[valid]) ]
best_p_row <- results_p[best_id, ]
best_p_row
```

```
##      p      avg_RMSE
## 7     7 0.003891943
```

```

p_opt <- best_p_row$p

var_ts <- df_subset %>%
  dplyr::select(all_of(vars_3)) %>%
  ts(start = c(start_year, start_month), frequency = 12) %>%
  stats::na.omit()

var_base <- VAR(y = var_ts, p = p_opt, type = "const")
summary(var_base)

```

```

##
## VAR Estimation Results:
## =====
## Endogenous variables: GDP_per_w, NHOUSE_P_CAN, CPI_ALL_CAN
## Deterministic variables: const
## Sample size: 528
## Log Likelihood: 6711.085
## Roots of the characteristic polynomial:
## 0.9211 0.7783 0.7783 0.7319 0.7319 0.7227 0.7109 0.7109 0.703 0.6649 0.6649 0.5845 0.5845 0.5688 0.5688
## Call:
## VAR(y = var_ts, p = p_opt, type = "const")
##
##
## Estimation results for equation GDP_per_w:
## =====
## GDP_per_w = GDP_per_w.l1 + NHOUSE_P_CAN.l1 + CPI_ALL_CAN.l1 + GDP_per_w.l2 + NHOUSE_P_CAN.l2 + CPI_ALL_CAN.l2 +
##
##              Estimate Std. Error t value Pr(>|t|)
## GDP_per_w.l1   -0.1788319   0.0445082  -4.018 6.76e-05 ***
## NHOUSE_P_CAN.l1 -0.0619824   0.0590253  -1.050  0.29417
## CPI_ALL_CAN.l1  -0.0746054   0.0500381  -1.491  0.13659
## GDP_per_w.l2    -0.0049528   0.0453108  -0.109  0.91300
## NHOUSE_P_CAN.l2  0.0478663   0.0639724   0.748  0.45467
## CPI_ALL_CAN.l2   0.0084386   0.0509387   0.166  0.86849
## GDP_per_w.l3     0.1033891   0.0453841   2.278  0.02314 *
## NHOUSE_P_CAN.l3 -0.0167907   0.0649799  -0.258  0.79620
## CPI_ALL_CAN.l3   0.1226546   0.0510725   2.402  0.01668 *
## GDP_per_w.l4    -0.1186841   0.0456887  -2.598  0.00966 **
## NHOUSE_P_CAN.l4  0.0189838   0.0650342   0.292  0.77048
## CPI_ALL_CAN.l4  -0.0119571   0.0515239  -0.232  0.81658
## GDP_per_w.l5    -0.0319407   0.0454405  -0.703  0.48243
## NHOUSE_P_CAN.l5  0.0521767   0.0648803   0.804  0.42166
## CPI_ALL_CAN.l5  -0.0455983   0.0512189  -0.890  0.37375
## GDP_per_w.l6     0.0220586   0.0452705   0.487  0.62628
## NHOUSE_P_CAN.l6 -0.1176570   0.0627399  -1.875  0.06133 .
## CPI_ALL_CAN.l6   0.0275494   0.0518533   0.531  0.59545
## GDP_per_w.l7     0.0608739   0.0438273   1.389  0.16546
## NHOUSE_P_CAN.l7  0.0087175   0.0568327   0.153  0.87815
## CPI_ALL_CAN.l7  -0.0316174   0.0502827  -0.629  0.52977
## const           0.0009825   0.0003033   3.240  0.00127 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

```

```

##
## Residual standard error: 0.004092 on 506 degrees of freedom
## Multiple R-Squared: 0.09877, Adjusted R-squared: 0.06137
## F-statistic: 2.641 on 21 and 506 DF, p-value: 0.0001128
##
##
## Estimation results for equation NHOUSE_P_CAN:
## =====
## NHOUSE_P_CAN = GDP_per_w.l1 + NHOUSE_P_CAN.l1 + CPI_ALL_CAN.l1 + GDP_per_w.l2 + NHOUSE_P_CAN.l2 + CPI
##
##           Estimate Std. Error t value Pr(>|t|)
## GDP_per_w.l1      0.0817191  0.0334698   2.442  0.01496 *
## NHOUSE_P_CAN.l1   0.4440717  0.0443865  10.005 < 2e-16 ***
## CPI_ALL_CAN.l1    -0.0355207  0.0376282  -0.944  0.34563
## GDP_per_w.l2      0.0860595  0.0340733   2.526  0.01185 *
## NHOUSE_P_CAN.l2   0.2586089  0.0481067   5.376 1.17e-07 ***
## CPI_ALL_CAN.l2     0.0769650  0.0383054   2.009  0.04504 *
## GDP_per_w.l3      0.0999473  0.0341285   2.929  0.00356 **
## NHOUSE_P_CAN.l3    0.0152934  0.0488643   0.313  0.75443
## CPI_ALL_CAN.l3    -0.0896145  0.0384061  -2.333  0.02002 *
## GDP_per_w.l4      0.0076720  0.0343575   0.223  0.82339
## NHOUSE_P_CAN.l4   -0.0118946  0.0489052  -0.243  0.80794
## CPI_ALL_CAN.l4     0.0478222  0.0387455   1.234  0.21768
## GDP_per_w.l5     -0.0046991  0.0341708  -0.138  0.89068
## NHOUSE_P_CAN.l5    0.0604948  0.0487895   1.240  0.21558
## CPI_ALL_CAN.l5    -0.0962581  0.0385162  -2.499  0.01276 *
## GDP_per_w.l6     -0.0686208  0.0340430  -2.016  0.04436 *
## NHOUSE_P_CAN.l6    0.1142729  0.0471798   2.422  0.01578 *
## CPI_ALL_CAN.l6    -0.0594715  0.0389932  -1.525  0.12784
## GDP_per_w.l7     -0.0327846  0.0329577  -0.995  0.32034
## NHOUSE_P_CAN.l7   -0.0203373  0.0427377  -0.476  0.63438
## CPI_ALL_CAN.l7    -0.0008801  0.0378122  -0.023  0.98144
## const             0.0005143  0.0002281   2.255  0.02455 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.003077 on 506 degrees of freedom
## Multiple R-Squared: 0.5946, Adjusted R-squared: 0.5778
## F-statistic: 35.34 on 21 and 506 DF, p-value: < 2.2e-16
##
##
## Estimation results for equation CPI_ALL_CAN:
## =====
## CPI_ALL_CAN = GDP_per_w.l1 + NHOUSE_P_CAN.l1 + CPI_ALL_CAN.l1 + GDP_per_w.l2 + NHOUSE_P_CAN.l2 + CPI
##
##           Estimate Std. Error t value Pr(>|t|)
## GDP_per_w.l1      0.0031934  0.0395476   0.081  0.9357
## NHOUSE_P_CAN.l1   0.0190652  0.0524467   0.364  0.7164
## CPI_ALL_CAN.l1    0.2095873  0.0444612   4.714 3.15e-06 ***
## GDP_per_w.l2      0.0089781  0.0402607   0.223  0.8236
## NHOUSE_P_CAN.l2  -0.0305304  0.0568425  -0.537  0.5914
## CPI_ALL_CAN.l2     0.0287840  0.0452614   0.636  0.5251
## GDP_per_w.l3      0.0115723  0.0403259   0.287  0.7743

```

```

## NHOUSE_P_CAN.13  0.1087477  0.0577377  1.883  0.0602 .
## CPI_ALL_CAN.13   0.0901489  0.0453803  1.987  0.0475 *
## GDP_per_w.14     -0.0400105  0.0405965 -0.986  0.3248
## NHOUSE_P_CAN.14  -0.1007248  0.0577859 -1.743  0.0819 .
## CPI_ALL_CAN.14   0.0905948  0.0457813  1.979  0.0484 *
## GDP_per_w.15     -0.0526975  0.0403759 -1.305  0.1924
## NHOUSE_P_CAN.15  0.0191373  0.0576492  0.332  0.7401
## CPI_ALL_CAN.15   -0.0476408  0.0455104 -1.047  0.2957
## GDP_per_w.16     -0.0288366  0.0402249 -0.717  0.4738
## NHOUSE_P_CAN.16  0.0647782  0.0557473  1.162  0.2458
## CPI_ALL_CAN.16   0.0006603  0.0460740  0.014  0.9886
## GDP_per_w.17     -0.0091664  0.0389426 -0.235  0.8140
## NHOUSE_P_CAN.17  -0.0516329  0.0504985 -1.022  0.3071
## CPI_ALL_CAN.17   -0.0004073  0.0446785 -0.009  0.9927
## const            0.0014280  0.0002695  5.299 1.74e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.003636 on 506 degrees of freedom
## Multiple R-Squared: 0.09271, Adjusted R-squared: 0.05505
## F-statistic: 2.462 on 21 and 506 DF, p-value: 0.0003486
##
##
## Covariance matrix of residuals:
##           GDP_per_w NHOUSE_P_CAN CPI_ALL_CAN
## GDP_per_w  1.674e-05  9.212e-07  7.155e-07
## NHOUSE_P_CAN 9.212e-07  9.468e-06  8.139e-08
## CPI_ALL_CAN  7.155e-07  8.139e-08  1.322e-05
##
## Correlation matrix of residuals:
##           GDP_per_w NHOUSE_P_CAN CPI_ALL_CAN
## GDP_per_w  1.00000  0.073163  0.048095
## NHOUSE_P_CAN 0.07316  1.000000  0.007276
## CPI_ALL_CAN  0.04809  0.007276  1.000000

```

2.3 Small VAR: Long term Model (h=1-12)

```

results_pl <- data.frame(p = integer(), avg_RMSE = numeric())

for (p in p_grid) {
  avg_rmse1 <- ts_cv_rmse_for_p(X, p, K = 10, max_h = 12) # CV over horizons 1-3
  results_pl <- rbind(results_pl, data.frame(p = p, avg_RMSE = avg_rmse1))
  cat("Done CV: p =", p, "avg_RMSE =", round(avg_rmse1, 6), "\n")
}

## Done CV: p = 1 avg_RMSE = 0.004263
## Done CV: p = 2 avg_RMSE = 0.004383
## Done CV: p = 3 avg_RMSE = 0.004113
## Done CV: p = 4 avg_RMSE = 0.004147
## Done CV: p = 5 avg_RMSE = 0.004772

```

```
## Done CV: p = 6 avg_RMSE = 0.006166
## Done CV: p = 7 avg_RMSE = 0.003624
## Done CV: p = 8 avg_RMSE = 0.003738
## Done CV: p = 9 avg_RMSE = 0.003823
## Done CV: p = 10 avg_RMSE = 0.003777
## Done CV: p = 11 avg_RMSE = 0.00391
## Done CV: p = 12 avg_RMSE = 0.004035
```

```
print(results_pl)
```

```
##      p      avg_RMSE
## 1     1 0.004262744
## 2     2 0.004383386
## 3     3 0.004112786
## 4     4 0.004146604
## 5     5 0.004771585
## 6     6 0.006165701
## 7     7 0.003623924
## 8     8 0.003738310
## 9     9 0.003823238
## 10    10 0.003777167
## 11    11 0.003910353
## 12    12 0.004035132
```

```
valid1      <- which(is.finite(results_pl$avg_RMSE))
best_id1    <- valid1[ which.min(results_pl$avg_RMSE[valid1]) ]
best_p_row1 <- results_pl[best_id1, ]
best_p_row1
```

```
##      p      avg_RMSE
## 7 7 0.003623924
```

```
p_opt1 <- best_p_row1$p
```

```
var_base <- VAR(y = var_ts, p = p_opt1, type = "const")
summary(var_base)
```

```
##
## VAR Estimation Results:
## =====
## Endogenous variables: GDP_per_w, NHOUSE_P_CAN, CPI_ALL_CAN
## Deterministic variables: const
## Sample size: 528
## Log Likelihood: 6711.085
## Roots of the characteristic polynomial:
## 0.9211 0.7783 0.7783 0.7319 0.7319 0.7227 0.7109 0.7109 0.703 0.6649 0.6649 0.5845 0.5845 0.5688 0.5
## Call:
## VAR(y = var_ts, p = p_opt1, type = "const")
##
##
## Estimation results for equation GDP_per_w:
## =====
```

```

## GDP_per_w = GDP_per_w.l1 + NHOUSE_P_CAN.l1 + CPI_ALL_CAN.l1 + GDP_per_w.l2 + NHOUSE_P_CAN.l2 + CPI_A
##
##
##           Estimate Std. Error t value Pr(>|t|)
## GDP_per_w.l1      -0.1788319  0.0445082  -4.018 6.76e-05 ***
## NHOUSE_P_CAN.l1   -0.0619824  0.0590253  -1.050  0.29417
## CPI_ALL_CAN.l1    -0.0746054  0.0500381  -1.491  0.13659
## GDP_per_w.l2      -0.0049528  0.0453108  -0.109  0.91300
## NHOUSE_P_CAN.l2    0.0478663  0.0639724   0.748  0.45467
## CPI_ALL_CAN.l2     0.0084386  0.0509387   0.166  0.86849
## GDP_per_w.l3       0.1033891  0.0453841   2.278  0.02314 *
## NHOUSE_P_CAN.l3   -0.0167907  0.0649799  -0.258  0.79620
## CPI_ALL_CAN.l3     0.1226546  0.0510725   2.402  0.01668 *
## GDP_per_w.l4      -0.1186841  0.0456887  -2.598  0.00966 **
## NHOUSE_P_CAN.l4    0.0189838  0.0650342   0.292  0.77048
## CPI_ALL_CAN.l4    -0.0119571  0.0515239  -0.232  0.81658
## GDP_per_w.l5      -0.0319407  0.0454405  -0.703  0.48243
## NHOUSE_P_CAN.l5    0.0521767  0.0648803   0.804  0.42166
## CPI_ALL_CAN.l5    -0.0455983  0.0512189  -0.890  0.37375
## GDP_per_w.l6       0.0220586  0.0452705   0.487  0.62628
## NHOUSE_P_CAN.l6   -0.1176570  0.0627399  -1.875  0.06133 .
## CPI_ALL_CAN.l6     0.0275494  0.0518533   0.531  0.59545
## GDP_per_w.l7       0.0608739  0.0438273   1.389  0.16546
## NHOUSE_P_CAN.l7    0.0087175  0.0568327   0.153  0.87815
## CPI_ALL_CAN.l7    -0.0316174  0.0502827  -0.629  0.52977
## const              0.0009825  0.0003033   3.240  0.00127 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.004092 on 506 degrees of freedom
## Multiple R-Squared:  0.09877, Adjusted R-squared:  0.06137
## F-statistic: 2.641 on 21 and 506 DF,  p-value: 0.0001128
##
##
## Estimation results for equation NHOUSE_P_CAN:
## =====
## NHOUSE_P_CAN = GDP_per_w.l1 + NHOUSE_P_CAN.l1 + CPI_ALL_CAN.l1 + GDP_per_w.l2 + NHOUSE_P_CAN.l2 + CP
##
##
##           Estimate Std. Error t value Pr(>|t|)
## GDP_per_w.l1       0.0817191  0.0334698   2.442  0.01496 *
## NHOUSE_P_CAN.l1    0.4440717  0.0443865  10.005 < 2e-16 ***
## CPI_ALL_CAN.l1    -0.0355207  0.0376282  -0.944  0.34563
## GDP_per_w.l2       0.0860595  0.0340733   2.526  0.01185 *
## NHOUSE_P_CAN.l2    0.2586089  0.0481067   5.376 1.17e-07 ***
## CPI_ALL_CAN.l2     0.0769650  0.0383054   2.009  0.04504 *
## GDP_per_w.l3       0.0999473  0.0341285   2.929  0.00356 **
## NHOUSE_P_CAN.l3    0.0152934  0.0488643   0.313  0.75443
## CPI_ALL_CAN.l3    -0.0896145  0.0384061  -2.333  0.02002 *
## GDP_per_w.l4       0.0076720  0.0343575   0.223  0.82339
## NHOUSE_P_CAN.l4   -0.0118946  0.0489052  -0.243  0.80794
## CPI_ALL_CAN.l4     0.0478222  0.0387455   1.234  0.21768
## GDP_per_w.l5      -0.0046991  0.0341708  -0.138  0.89068
## NHOUSE_P_CAN.l5    0.0604948  0.0487895   1.240  0.21558
## CPI_ALL_CAN.l5    -0.0962581  0.0385162  -2.499  0.01276 *

```

```

## GDP_per_w.16      -0.0686208  0.0340430  -2.016  0.04436 *
## NHOUSE_P_CAN.16   0.1142729  0.0471798   2.422  0.01578 *
## CPI_ALL_CAN.16    -0.0594715  0.0389932  -1.525  0.12784
## GDP_per_w.17      -0.0327846  0.0329577  -0.995  0.32034
## NHOUSE_P_CAN.17   -0.0203373  0.0427377  -0.476  0.63438
## CPI_ALL_CAN.17    -0.0008801  0.0378122  -0.023  0.98144
## const              0.0005143  0.0002281   2.255  0.02455 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.003077 on 506 degrees of freedom
## Multiple R-Squared: 0.5946, Adjusted R-squared: 0.5778
## F-statistic: 35.34 on 21 and 506 DF, p-value: < 2.2e-16
##
##
## Estimation results for equation CPI_ALL_CAN:
## =====
## CPI_ALL_CAN = GDP_per_w.11 + NHOUSE_P_CAN.11 + CPI_ALL_CAN.11 + GDP_per_w.12 + NHOUSE_P_CAN.12 + CPI
##
##              Estimate Std. Error t value Pr(>|t|)
## GDP_per_w.11    0.0031934  0.0395476   0.081   0.9357
## NHOUSE_P_CAN.11 0.0190652  0.0524467   0.364   0.7164
## CPI_ALL_CAN.11   0.2095873  0.0444612   4.714 3.15e-06 ***
## GDP_per_w.12     0.0089781  0.0402607   0.223   0.8236
## NHOUSE_P_CAN.12 -0.0305304  0.0568425  -0.537   0.5914
## CPI_ALL_CAN.12   0.0287840  0.0452614   0.636   0.5251
## GDP_per_w.13     0.0115723  0.0403259   0.287   0.7743
## NHOUSE_P_CAN.13  0.1087477  0.0577377   1.883   0.0602 .
## CPI_ALL_CAN.13   0.0901489  0.0453803   1.987   0.0475 *
## GDP_per_w.14     -0.0400105  0.0405965  -0.986   0.3248
## NHOUSE_P_CAN.14 -0.1007248  0.0577859  -1.743   0.0819 .
## CPI_ALL_CAN.14   0.0905948  0.0457813   1.979   0.0484 *
## GDP_per_w.15     -0.0526975  0.0403759  -1.305   0.1924
## NHOUSE_P_CAN.15  0.0191373  0.0576492   0.332   0.7401
## CPI_ALL_CAN.15   -0.0476408  0.0455104  -1.047   0.2957
## GDP_per_w.16     -0.0288366  0.0402249  -0.717   0.4738
## NHOUSE_P_CAN.16  0.0647782  0.0557473   1.162   0.2458
## CPI_ALL_CAN.16   0.0006603  0.0460740   0.014   0.9886
## GDP_per_w.17     -0.0091664  0.0389426  -0.235   0.8140
## NHOUSE_P_CAN.17 -0.0516329  0.0504985  -1.022   0.3071
## CPI_ALL_CAN.17   -0.0004073  0.0446785  -0.009   0.9927
## const            0.0014280  0.0002695   5.299 1.74e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.003636 on 506 degrees of freedom
## Multiple R-Squared: 0.09271, Adjusted R-squared: 0.05505
## F-statistic: 2.462 on 21 and 506 DF, p-value: 0.0003486
##
##
##
## Covariance matrix of residuals:

```

```
##          GDP_per_w NHOUSE_P_CAN CPI_ALL_CAN
## GDP_per_w  1.674e-05  9.212e-07  7.155e-07
## NHOUSE_P_CAN 9.212e-07  9.468e-06  8.139e-08
## CPI_ALL_CAN  7.155e-07  8.139e-08  1.322e-05
##
## Correlation matrix of residuals:
##          GDP_per_w NHOUSE_P_CAN CPI_ALL_CAN
## GDP_per_w  1.00000  0.073163  0.048095
## NHOUSE_P_CAN 0.07316  1.000000  0.007276
## CPI_ALL_CAN  0.04809  0.007276  1.000000
```

3. Large model Variable selection: How we select the FINAL 17

```
targets <- c("GDP_per_w", "CPI_ALL_CAN", "NHOUSE_P_CAN")

all_numeric <- names(df_large)[sapply(df_large, is.numeric)]
all_numeric <- setdiff(all_numeric, "Date")

## 1. Correlation scores vs targets
corr_score <- function(vname) {
  x <- df_large[[vname]]
  sapply(targets, function(tn) {
    y <- df_large[[tn]]
    abs(cor(x, y, use = "complete.obs"))
  }) |>
  max(na.rm = TRUE)
}

# score for variable
scores_vec <- sapply(all_numeric, corr_score)

corr_table <- data.frame(
  variable = all_numeric,
  score    = as.numeric(scores_vec)
) |>
  dplyr::arrange(dplyr::desc(score))

# Keep top 40 by corr scores
corr_table <- corr_table[!corr_table$variable %in% targets, ]
top40_vars <- head(corr_table$variable, 40)

print(head(corr_table, 40))
```

```
##          variable      score
## 4          IP_new 0.48780069
## 5          GPI_new 0.47433802
## 6          BSI_new 0.44706386
## 7          GDP_new 0.43577834
## 8          NDM_new 0.36619648
## 9          DM_new 0.34514662
## 10         OILP_new 0.34415978
```

```

## 11          SPI_new 0.33817485
## 12          WT_new 0.32939681
## 13          RT_new 0.29583675
## 14          WTISPLC 0.28131776
## 15          Exp_BP_new 0.26938742
## 16          CRED_HOUS 0.23885055
## 17          CRED_HOUS_MORT 0.23122263
## 18          G_AVG_10p.TBILL_3M 0.22604070
## 19          BANK_RATE_L 0.21849942
## 20          Imp_BP_new 0.20924419
## 21          FIN_new 0.20825877
## 22          IPPI_METAL_CAN 0.20387118
## 23          G_AVG_5.10.Bank_rate 0.19575219
## 24          TBILL_3M 0.19258680
## 25          CON_new 0.18541867
## 26          TBILL_6M 0.18404598
## 27          UNEMP_CAN 0.17534409
## 28          EMP_CONS_CAN 0.16654923
## 29          EMP_PART_CAN 0.13952844
## 30          EMP_FOR_OIL_CAN 0.13682628
## 31          UNEMP_DURA_5.13_CAN 0.13393978
## 32          EMP_SERV_CAN 0.13151818
## 33          EMP_MANU_CAN 0.13030941
## 34          EMP_CAN 0.12180263
## 35          hstart_CAN_new 0.12173111
## 36          USDCAD_new 0.12024532
## 37          TSX_CLO 0.10885666
## 38          TOT_HRS_CAN 0.10839831
## 39          OIL_CAN_new 0.10428180
## 40          TSX_HI 0.10205373
## 41          EMP_SALES_CAN 0.09967414
## 42          PA_new 0.08935124
## 43          EMP_FIN_CAN 0.08754307

```

```

## ----- separation line
## 2. Define economic blocks

```

```

block_real <- c(
  "GDP_new", "BSI_new", "GPI_new", "SPI_new",
  "IP_new", "NDM_new", "DM_new", "CON_new",
  "RT_new", "WT_new", "PA_new", "FIN_new",
  "TOT_HRS_CAN", "GOOD_HRS_CAN"
)

block_labour <- c(
  "EMP_CAN", "EMP_SERV_CAN", "EMP_FOR_OIL_CAN",
  "EMP_CONS_CAN", "EMP_SALES_CAN", "EMP_FIN_CAN",
  "EMP_MANU_CAN", "EMP_PART_CAN",
  "UNEMP_CAN",
  "UNEMP_DURA_1-4_CAN", "UNEMP_DURA_5-13_CAN",
  "UNEMP_DURA_14-25_CAN", "UNEMP_DURA_27+_CAN",
  "UNEMP_DURAvg_CAN_new",
  "CLAIMS_CAN",
  "TOT_HRS_CAN", "GOOD_OVT_HRS_CAN"
)

```

```

)

block_house_price <- c(
  "NHOUSE_P_CAN",
  "NHOUSE_P_NF", "NHOUSE_P_PEI", "NHOUSE_P_NS", "NHOUSE_P_NB",
  "NHOUSE_P_QC", "NHOUSE_P_ONT", "NHOUSE_P_MAN", "NHOUSE_P_SAS",
  "NHOUSE_P_ALB", "NHOUSE_P_BC"
)

block_house_activity <- c(
  "hstart_CAN_new",
  "hstart_NF_new", "hstart_PEI_new", "hstart_NS_new",
  "hstart_NB_new", "hstart_QC_new", "hstart_ONT_new",
  "hstart_MAN_new", "hstart_SAS_new", "hstart_ALB_new",
  "hstart_BC_new",
  "build_Total_CAN_new", "build_Ind_CAN_new", "build_Comm_CAN_new",
  "build_Total_NF_new", "build_Total_PEI_new", "build_Total_NS_new",
  "build_Total_NB_new", "build_Total_QC_new", "build_Total_ONT_new",
  "build_Total_MAN_new", "build_Total_SAS_new", "build_Total_ALB_new",
  "build_Total_BC_new"
)

block_manu <- c(
  "MANU_N_ORD_new", "MANU_UNFIL_new", "MANU_TOT_INV_new",
  "MANU_INV_RAT_new", "N_DUR_INV_RAT_new",
  "DUR_N_ORD_new", "DUR_UNFIL_new",
  "DUR_TOT_INV_new", "DUR_INV_RAT_new"
)

block_money_credit <- c(
  "M3", "M2p", "M_BASE1",
  "CRED_BUS_cb", "CRED_HOUS_cb", "CRED_MORT_HOUSE_cb", "CRED_T_cb",
  "CRED_HOUS_non_MORT", "CRED_HOUS_MORT", "CRED_HOUS", "CRED_BUS"
)

block_rates <- c(
  "BANK_RATE_L",
  "GOV_AVG_1_3Y", "GOV_AVG_3_5Y", "GOV_AVG_5_10Y", "GOV_AVG_10pY",
  "MORTG_1Y", "MORTG_5Y",
  "TBILL_3M", "TBILL_6M",
  "G_AVG_1-3-Bank_rate", "G_AVG_3-5-Bank_rate",
  "G_AVG_5-10-Bank_rate", "TBILL_6M-Bank_rate",
  "G_AVG_10p-TBILL_3M"
)

block_external <- c(
  "RES_TOT", "RES_USD", "RES_IMF",
  "Imp_BP_new", "IOIL_BP_new", "Exp_BP_new", "EOIL_BP_new",
  "EX_ENER_BP_new", "EX_MINER_BP_new", "EX_METAL_BP_new",
  "EX_IND_EQUIP_BP_new", "EX_TRANSP_BP_new", "EX_CONS_BP_new",
  "IMP_METAL_BP_new", "IMP_IND_EQUIP_BP_new",
  "IMP_TRANSP_BP_new", "IMP_CONS_BP_new",
  "USDCAD_new", "JPYCAD_new", "GBPCAD_new",

```

```

"CAN_EQTY_NETFLOW", "CAN_SEC_NETFLOW",
"FOR_SEC_NETFLOW", "CAN_US_SEC_NETFLOW"
)

block_prices <- c(
  "CPI_ALL_CAN", "CPI_SHEL_CAN", "CPI_CLOT_CAN", "CPI_HEA_CAN",
  "CPI_MINUS_FOO_CAN", "CPI_MINUS_FEN_CAN",
  "CPI_GOO_CAN", "CPI_DUR_CAN", "CPI_SERV_CAN",
  "IPPI_CAN", "IPPI_ENER_CAN", "IPPI_WOOD_CAN",
  "IPPI_METAL_CAN", "IPPI_MOTOR_CAN", "IPPI_MACH_CAN"
)

block_markets <- c(
  "OILP_new", "OIL_CAN_new", "WTISPLC",
  "TSX_HI", "TSX_LO", "TSX_CLO"
)

blocks <- list(
  real      = block_real,
  labour    = block_labour,
  house_price = block_house_price,
  house_activity = block_house_activity,
  manu      = block_manu,
  money_credit = block_money_credit,
  rates     = block_rates,
  external  = block_external,
  prices    = block_prices,
  markets   = block_markets
)

## 3. Helper: score lookup + within-block selection

score_lookup <- function(v) {
  out <- corr_table$score[match(v, corr_table$variable)]
  ifelse(is.na(out), 0, out)
}

pick_block_reps <- function(block_vars,
                             top_vars,
                             df,
                             n_per_block = 2,
                             corr_cutoff = 0.90) {
  cand <- intersect(block_vars, intersect(top_vars, names(df)))
  if (length(cand) == 0) return(character(0))

  cand_scores <- score_lookup(cand)
  ord <- order(cand_scores, decreasing = TRUE)
  cand <- cand[ord]

  picked <- character(0)

  for (v in cand) {
    if (length(picked) == 0) {

```

```

    picked <- c(picked, v)
  } else {
    too_close <- FALSE
    for (u in picked) {
      cval <- cor(df[[v]], df[[u]], use = "complete.obs")
      if (is.finite(cval) && abs(cval) > corr_cutoff) {
        too_close <- TRUE
        break
      }
    }
    if (!too_close) {
      picked <- c(picked, v)
    }
  }
  if (length(picked) >= n_per_block) break
}

picked
}

## 4. Apply: choose up to 2 per block (variety)

set.seed(12345)

selected_by_block <- lapply(names(blocks), function(bn) {
  vars <- pick_block_reps(
    block_vars = blocks[[bn]],
    top_vars    = top40_vars,
    df          = df_large,
    n_per_block = 2,
    corr_cutoff = 0.90
  )

  if (length(vars) == 0) return(NULL)

  data.frame(
    block      = bn,
    variable   = vars,
    stringsAsFactors = FALSE
  )
})

selected_by_block <- do.call(rbind, selected_by_block)
selected_by_block

```

```

##           block      variable
## 1          real      IP_new
## 2          real      BSI_new
## 3        labour    UNEMP_CAN
## 4        labour    EMP_CONS_CAN
## 5 house_activity hstart_CAN_new
## 6   money_credit    CRED_HOUS

```

```
## 7    money_credit CRED_HOUS_MORT
## 8      rates     BANK_RATE_L
## 9      rates     TBILL_3M
## 10    external   Exp_BP_new
## 11    external   Imp_BP_new
## 12      prices IPPI_METAL_CAN
## 13      markets   OILP_new
## 14      markets   WTISPLC
```

5. Final large-model variable set

```
vars_large_final <- unique(c(
  targets,
  selected_by_block$variable
))

length(vars_large_final)
```

```
## [1] 17
```

```
vars_large_final
```

```
## [1] "GDP_per_w"      "CPI_ALL_CAN"    "NHOUSE_P_CAN"   "IP_new"
## [5] "BSI_new"        "UNEMP_CAN"      "EMP_CONS_CAN"   "hstart_CAN_new"
## [9] "CRED_HOUS"      "CRED_HOUS_MORT" "BANK_RATE_L"    "TBILL_3M"
## [13] "Exp_BP_new"     "Imp_BP_new"     "IPPI_METAL_CAN" "OILP_new"
## [17] "WTISPLC"
```

4. Large Model: Short Horizon

```
X_large <- df_large %>%
  dplyr::select(all_of(vars_large_final)) %>%
  as.matrix() %>% na.omit()

p_grid_short <- 1:12
ts_cv_rmse_TARGETS_only <- function(X, p, K = 10, max_h = 3, target_idx) {
  n <- nrow(X)
  start_cv <- floor(n * 0.4)
  cut_points <- round(seq(start_cv, n - max_h, length.out = K))

  err_list <- list()

  for (t0 in cut_points) {
    train <- X[1:t0, , drop = FALSE]
    actual <- X[(t0+1):(t0+max_h), target_idx, drop = FALSE]

    model <- try(VAR(train, p = p, type = "const"), silent = TRUE)
    if (inherits(model, "try-error")) next

    pred_list <- predict(model, n.ahead = max_h)$fcst
```

```

    pred_mat <- sapply(target_idx, function(j) pred_list[[j]][,"fcst"])

    err <- actual - pred_mat
    err_list[[length(err_list)+1]] <- err
  }

  all_err <- do.call(rbind, err_list)
  sqrt(mean(all_err^2, na.rm = TRUE))
}

target_idx <- match(
  c("GDP_per_w", "CPI_ALL_CAN", "NHOUSE_P_CAN"),
  vars_large_final
)

results_large_short_opt2 <- data.frame(p = integer(), avg_RMSE = numeric())

for (p in p_grid_short) {
  rmse_p <- ts_cv_rmse_TARGETS_only(
    X = X_large,
    p = p,
    K = 10,
    max_h = 3,
    target_idx = target_idx
  )
  results_large_short_opt2 <- rbind(
    results_large_short_opt2,
    data.frame(p = p, avg_RMSE = rmse_p)
  )
  cat("Large VAR Short CV: p =", p, "| RMSE =", rmse_p, "\n")
}

```

```

## Large VAR Short CV: p = 1 | RMSE = 0.003154988
## Large VAR Short CV: p = 2 | RMSE = 0.003000834
## Large VAR Short CV: p = 3 | RMSE = 0.003142708
## Large VAR Short CV: p = 4 | RMSE = 0.003197658
## Large VAR Short CV: p = 5 | RMSE = 0.003494847
## Large VAR Short CV: p = 6 | RMSE = 0.004098532
## Large VAR Short CV: p = 7 | RMSE = 0.004483536
## Large VAR Short CV: p = 8 | RMSE = 0.004088858
## Large VAR Short CV: p = 9 | RMSE = 0.00414951
## Large VAR Short CV: p = 10 | RMSE = 0.004918617
## Large VAR Short CV: p = 11 | RMSE = 0.005630393
## Large VAR Short CV: p = 12 | RMSE = 0.005034551

```

```
## it give CV table p=[1:12] lol
```

```

best_id2 <- which.min(results_large_short_opt2$avg_RMSE)
best_p_large_short_opt2 <- results_large_short_opt2$p[best_id2]
best_p_large_short_opt2

```

```
## [1] 2
```

5. Large: model: Long Horizon

```
p_grid_long <- 1:12
results_large_long_opt2 <- data.frame(p = integer(), avg_RMSE = numeric())

for (p in p_grid_long) {
  rmse_p <- ts_cv_rmse_TARGETS_only(
    X = X_large,
    p = p,
    K = 10,
    max_h = 12,
    target_idx = target_idx
  )
  results_large_long_opt2 <- rbind(
    results_large_long_opt2,
    data.frame(p = p, avg_RMSE = rmse_p)
  )
  cat("Large VAR Long CV: p =", p, "| RMSE =", rmse_p, "\n")
}
```

```
## Large VAR Long CV: p = 1 | RMSE = 0.003423545
## Large VAR Long CV: p = 2 | RMSE = 0.003443184
## Large VAR Long CV: p = 3 | RMSE = 0.003514326
## Large VAR Long CV: p = 4 | RMSE = 0.003690847
## Large VAR Long CV: p = 5 | RMSE = 0.003965697
## Large VAR Long CV: p = 6 | RMSE = 0.004224772
## Large VAR Long CV: p = 7 | RMSE = 0.004609542
## Large VAR Long CV: p = 8 | RMSE = 0.004268616
## Large VAR Long CV: p = 9 | RMSE = 0.004438562
## Large VAR Long CV: p = 10 | RMSE = 0.005668446
## Large VAR Long CV: p = 11 | RMSE = 0.006925354
## Large VAR Long CV: p = 12 | RMSE = 0.005854548
```

```
## it give CV table p=[1:12] lol
best_id2L <- which.min(results_large_long_opt2$avg_RMSE)
best_p_large_long_opt2 <- results_large_long_opt2$p[best_id2L]
best_p_large_long_opt2
```

```
## [1] 1
```

6. RMSE Comparison: Small VAR vs Large VAR

```
rmse_targets_only <- function(X, p, K = 10, max_h = 3, target_idx) {
  n <- nrow(X)
  start_cv <- floor(n * 0.4)
  cut_pts <- round(seq(start_cv, n - max_h, length.out = K))

  err_list <- list()
```

```

for (t0 in cut_pts) {
  train <- X[1:t0, , drop = FALSE]
  actual <- X[(t0 + 1):(t0 + max_h), target_idx, drop = FALSE]

  model <- try(VAR(train, p = p, type = "const"), silent = TRUE)
  if (inherits(model, "try-error")) next

  pred_obj <- predict(model, n.ahead = max_h)$fcst
  pred_mat <- sapply(target_idx, function(j) pred_obj[[j]][, "fcst"])

  err_list[[length(err_list) + 1]] <- actual - pred_mat
}

all_err <- do.call(rbind, err_list)
sqrt(mean(all_err^2, na.rm = TRUE))
}

## Short-term forecast performance (h = 1-3)
# Small model: RMSE from earlier K-fold CV (3-var VAR).
rmse_small_short <- min(results_p$avg_RMSE, na.rm = TRUE)

# Large model: RMSE for the same 3 target series
rmse_large_short_opt2 <- rmse_targets_only(
  X      = X_large,
  p      = best_p_large_short_opt2,
  K      = 10,
  max_h  = 3,
  target_idx = target_idx
)

short_term_table <- data.frame(
  Model      = c("Small VAR (3 variables)",
                 "Large VAR (17 variables)"),
  Lag_p      = c(p_opt,
                 best_p_large_short_opt2),
  RMSE_1to3  = c(rmse_small_short,
                 rmse_large_short_opt2)
)

short_term_table

##               Model Lag_p  RMSE_1to3
## 1 Small VAR (3 variables)    7 0.003891943
## 2 Large VAR (17 variables)    2 0.003000834

```

```

## Long-term forecast performance (h = 1-12)

# Small model: RMSE from small K-fold CV (3-variable VAR, long horizon)
rmse_small_long <- min(results_pl$avg_RMSE, na.rm = TRUE)

# Large model: RMSE of 3 target series
rmse_large_long_opt2 <- rmse_targets_only(
  X      = X_large,

```

```

p          = best_p_large_long_opt2,
K          = 10,
max_h      = 12,
target_idx = target_idx
)

long_term_table <- data.frame(
  Model      = c("Small VAR (3 variables)",
                 "Large VAR (17 variables)"),
  Lag_p      = c(p_opt1,
                 best_p_large_long_opt2),
  RMSE_1to12 = c(rmse_small_long,
                 rmse_large_long_opt2)
)

long_term_table

```

```

##                Model Lag_p  RMSE_1to12
## 1 Small VAR (3 variables)    7 0.003623924
## 2 Large VAR (17 variables)    1 0.003423545

```

7. Final baseline VAR models

```

var_ts_small <- ts(
  df_subset %>%
    dplyr::select(all_of(vars_3)),
  start      = c(start_year, start_month),
  frequency = 12
)

# Short-Horizon small VAR
var_small_short <- VAR(
  y      = var_ts_small,
  p      = p_opt,
  type   = "const"
)

# Long-Horizon small VAR
var_small_long <- VAR(
  y      = var_ts_small,
  p      = p_opt1,
  type   = "const"
)

var_ts_large <- ts(
  df_large %>%
    dplyr::select(all_of(vars_large_final)),
  start      = c(start_year, start_month),
  frequency = 12
)

# Short-term large VAR

```

```

var_large_short <- VAR(
  y = var_ts_large,
  p = best_p_large_short_opt2,
  type = "const"
)

# Long-term large VAR
var_large_long <- VAR(
  y = var_ts_large,
  p = best_p_large_long_opt2,
  type = "const"
)

## Too long hide it
#summary(var_small_short)
#summary(var_small_long)
#summary(var_large_short)
#summary(var_large_long)

```

1. 8.Forecast

```

H_short <- 3
H_long <- 12

# VAR forecasts
fc_large_short <- predict(var_large_short, n.ahead = H_short)$fcst
fc_large_long <- predict(var_large_long, n.ahead = H_long)$fcst

# Last observed date --> so no gap
last_date <- max(df_subset$Date)

dates_short <- seq(last_date %m+% months(1), by = "1 month", length.out = H_short)
dates_long <- seq(last_date %m+% months(1), by = "1 month", length.out = H_long)

make_var_long <- function(fc_obj, H, dates_vec) {
  tibble(
    variable = rep(c("GDP_per_w", "CPI_ALL_CAN", "NHOUSE_P_CAN"), each = H),
    h = rep(1:H, times = 3),
    date = rep(dates_vec, times = 3),
    forecast = c(
      fc_obj$GDP_per_w[, "fcst"],
      fc_obj$CPI_ALL_CAN[, "fcst"],
      fc_obj$NHOUSE_P_CAN[, "fcst"]
    )
  )
}

var_large_short_long <- make_var_long(fc_large_short, H_short, dates_short)
var_large_long_long <- make_var_long(fc_large_long, H_long, dates_long)

# CSV for the "empirical_results" file to merge other notebooks.
write.csv(var_large_short_long,
  "VAR_large_all_variables_short.csv",

```

```
    row.names = FALSE)  
write.csv(var_large_long_long,  
          "VAR_large_all_variables_long.csv",  
          row.names = FALSE)
```